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Precision Maternal Health: Robust High-Risk Pregnancy Prediction via Ensemble Learning and Explainable AI

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	prediction of high risk pregnancy with explainable artificial intelligence. The work has not been submitted elsewhere and complies with the journal ethical standards. We appreciate your consideration.

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Precision Maternal Health: Robust High-Risk Pregnancy Prediction via Ensemble Learning and Explainable AI

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Abstract

Purpose: High-risk pregnancies are one significant cause of maternal and fetal mortality in the world. Although diagnosis at the earliest stage is important, standard risk assessment methods usually depend on the subjective clinical evaluation. It is the purpose of this research to develop a high quality, data-driven machine learning (ML) model to make predictions of high-risk pregnancies.
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Conclusion: This framework balances between good predictive and clinical interpretability. It will be a trusted non-invasive decision-support tool by aligning predictions with known medical knowledge through XAI to instill confidence among obstetricians in clinical practice in practice.

Keywords: High-Risk Pregnancy, Machine Learning, Ensemble Learning, Explainable AI (XAI), Maternal Healthcare

1 Introduction

Maternal health remains a major global public health challenge, with pregnancy-related complications continuing to contribute substantially to maternal and neonatal morbidity and mortality. Early identification of high-risk pregnancies is essential for timely intervention and improved clinical outcomes; however, accurate prediction remains difficult due to the complex interactions among physiological, clinical, and demographic factors influencing maternal health [1]. Systematic evidence indicates that existing risk prediction models for maternal mortality often lack sufficient discrimination for individual clinical decision-making, highlighting the need for more reliable and data-driven approaches [1]. Advances in machine learning (ML) have introduced new opportunities for improving healthcare prediction systems by enabling automated analysis of complex clinical data and uncovering nonlinear relationships among risk factors [2]. In maternal healthcare, ML-based systems have been applied to predict fetal anomalies and pregnancy complications using maternal clinical history and physiological indicators, demonstrating promising predictive capabilities [2]. Nonetheless, pregnancy outcomes are inherently multifactorial, and traditional statistical models such as logistic regression frequently fail to capture intricate interactions among predictors [3].

Comparative studies have shown that advanced ML algorithms, including random forests and gradient boosting methods, can outperform conventional regression approaches in predicting pregnancy outcomes such as preterm delivery and gestational diabetes [3]. However, some population-based cohort studies report limited performance gains over traditional models, suggesting that predictive success depends heavily on dataset quality, feature engineering, and modeling strategies [4]. These findings underscore the importance of developing robust modeling frameworks that

effectively leverage diverse clinical features. Recent research has explored the use of ensemble learning and hybrid models to enhance predictive accuracy and generalization. Ensemble techniques combine multiple classifiers to exploit their complementary strengths, resulting in improved stability and performance in maternal risk prediction tasks [5]. For instance, ensemble frameworks integrating decision trees, boosting methods, and support vector machines have demonstrated improved classification accuracy for pregnancy risk levels [6]. Hybrid deep learning models that combine neural networks with tree-based classifiers have further improved performance, emphasizing the value of integrating heterogeneous learning paradigms [7].

In parallel, the integration of Explainable Artificial Intelligence (XAI) has emerged as a crucial advancement in clinical decision support systems. While high-performing ML models often operate as "black boxes," explainability techniques such as SHAP and LIME enable clinicians to interpret model predictions and understand feature contributions, thereby increasing trust and facilitating clinical adoption [8]. Explainable models have identified key physiological indicators—including blood pressure, glucose levels, and maternal age—as dominant risk factors, aligning model outputs with established medical knowledge [5, 8].

Numerous recent studies have reported strong predictive performance using ML techniques for maternal health risk classification. Tree-based models, gradient boosting, and support vector machines have achieved high accuracy in classifying risk levels using vital signs such as blood pressure, heart rate, and blood glucose [9, 10]. Ensemble frameworks and cross-validation strategies further enhance model reliability and support their integration into clinical decision support systems [11]. Moreover, large-scale datasets collected through IoT-enabled healthcare systems have enabled the development of high-performance predictive models suitable for deployment in resource-constrained settings [12].

Despite these advances, several challenges remain. Many studies rely on limited or region-specific datasets, restricting generalizability across populations [13]. Class imbalance and misclassification of intermediate risk categories continue to hinder model performance [14]. Additionally, the lack of interpretable and clinically transparent models remains a significant barrier to real-world adoption [15]. Addressing these challenges requires the development of robust, interpretable, and generalizable frameworks capable of delivering accurate predictions while providing clinically meaningful insights.

In this context, the present study proposes a precision maternal health framework for high-risk pregnancy prediction using ensemble learning and Explainable AI. By integrating multiple machine learning classifiers within a soft voting ensemble and employing SHAP-based interpretability, the proposed approach aims to enhance predictive accuracy, stability, and clinical transparency

The primary contributions of this study are:

- **Rigorous Comparative Evaluation:** A systematic assessment of five distinct ML algorithms utilizing Stratified 5-Fold Cross-Validation to ensure robust and generalizable results.

- **Stability-Optimized Ensemble:** The development of a Soft Voting Ensemble (integrating SVM, KNN, DT, and XGBoost) that achieves state-of-the-art accuracy (98.80%) while demonstrating superior stability (0.32% standard deviation) compared to standalone models.
- **Clinical Interpretability:** The application of SHAP analysis to validate that the model's feature prioritization aligns with established medical literature, ensuring the system's predictive logic is clinically sound.

2 Literature Review

Divergent work on different architectures has become a common trend in the application of Machine Learning in maternal healthcare with researchers experimenting with diverse architecture improvements to enhance risk stratification. This part is a review of the recent developments in the areas of general predictive modeling, ensemble learning approaches, and the newly developed area of Explainable AI (XAI) in obstetrics.

2.1 Machine Learning in Maternal Risk Prediction

The idea of machine learning as a means to forecast the risks of a mother is a relatively new one that is still being developed. The previous literature has affirmed the accuracy of traditional ML algorithms towards risk prediction of pregnancy through routine vital signs. The multi-layer Perceptron (MLP) and the Random Forest (RF) are some of the models applied by Pi et al. to the Bangladesh maternal health data [16]. They concluded that MLP was superior and superior to baselines but on standard vitals, simple algorithms like RF and Decision Trees (DT) compared well. Bearing the same statement, Al Mashrafi et al. [17] employed real life data of maternal death in Oman and concluded that the best tool to categorize the population in the case is the Random Forest. Uddin et al. also mention RF as the most recall effective in the context of the high-risk identification in a comparative assessment of the IoT-based healthcare models [18]. Such studies provide good baselines, but however, lay a lot of stress upon the roles of the maximization of headline accuracy criteria, without much regard to the variance and consistency of the models on diverse data splits that are significant in terms of clinical reliability.

2.2 Ensemble Learning and Model Stability

To overcome the disabilities of individual classifiers, the emphasis of recent literature has shifted towards ensemble techniques. Khadidos et al. [19] recommended a quad ensemble framework and demonstrated that the the integration of heterogeneous groups of learners improves the capacity of classifying data at the various risk levels of the multiple classes (low, medium, high). This is correlated with the findings by Malde et al. [20] who have emphasized the fact that ensemble methods would be particularly helpful in Lower-Middle-Income Countries (LMIC) where data is not only sparse but may also be noisy.

Specific to the voting system, Sidiqui and Hameed [21] explored a Soft Voting Ensemble in the risk of the mother, stating that probability based voting reflects the fine-tuning of the edges of complementary choices than hard voting does. We also assume this in our study but we go beyond this by presenting the intensive quantitative stability estimates through Stratified 5-Fold Cross-Validation. Not only do we give the mean accuracy but also the standard deviation of the method to show the stability of the Soft Voting method.

In addition, the dialogues have increased and declared that they can achieve a lot. The pipelines with XGBoost, often in combination with a dimensionality reduction (PCA) or imbalance correction (SMOTE) algorithms were employed by Sarker et al. [22] and Gao et al. to achieve high predictive performance [23]. It was also demonstrated by Liu et al. [24] that XGBoost can be applied in complex problems like forecasting bad events in Antiphospholipid Syndrome (APS). Our findings between xgboost and other single classifiers show that xgboost is the best choice but it also ends that the reduced variance in an ensemble should be viewed as an improved choice instead of a single boosting model in clinical practice.

2.3 Explainable AI (XAI) and Clinical Trust

As the degree of model complexity is on the rise, the black box problem is becoming a severe issue. It is increasingly emerging that a high degree of accuracy is not enough, unless it is interpretable. More recent works have started to apply the technique of XAI to validate the hypothesis of ML. An example uses of SHAP include the successful visualization of feature contribution to a hybrid fuzzy XGBoost model [25] and the use of SHAP to analyze postpartum depression literature to help clinicians gain insights on the risk factors.

Similarly, in another study, it is also found that the maternal continuum of care is susceptible to effective socio-demographic factors, which are identified by SHAP [26]. Mojumdar et al. [27] collected a collection of clinical data where the focus is on the risk factors like blood sugar, and blood pressure. Such clinical observations are connected to our practice and directly connected to behavior models. We can explicitly demonstrate the consistency of our models with medical domain knowledge since with SHAP of our ensemble and XGBoost models, we demonstrate that our algorithms give preferences to physiologically relevant competencies: Preexisting Diabetes, BMI and Blood Glucose.

All said and done, to the extent that literature has demonstrated that ML and ensembles may prove effective in maternal health, there remains a gap in enabling maximum predictive stability (when using low-variance ensembles) and interpretability (when using SHAP) simultaneously on typical clinical data. It is this gap in this paper that is addressed by giving the framework which is statistically strong and clinical explanation.

3 Methodology

This study has a systematic approach to high-risk pregnancy prediction, including the acquisition of data, serious preprocessing, the implementation of the classifier, its

assembly, and the interpretation of the results. The workflow proposal will guarantee high predictive and clinical interpretations. The entire architecture workflow of the proposed study is as shown in Figure 1

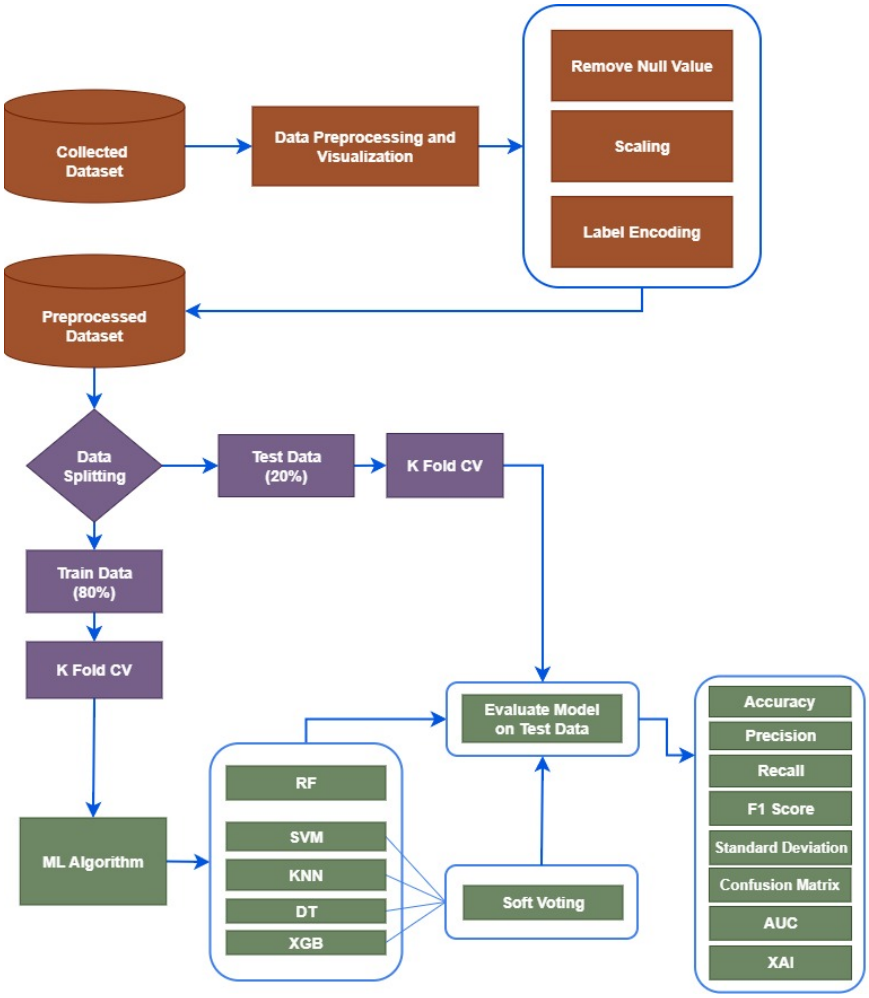

Fig. 1 Comprehensive flowchart of the proposed methodology.

The research starts with the maternal health data acquisition followed by a cleaning stage to cover any missing data (e.g., in BMI) and normalizing numerical variables as shown in Figure 1. The training phase will entail training five base classifiers and combining their predictions using the Soft Voting Ensemble. The last stage is devoted to validation and explainability, which is necessary to acquire the stable and clinically transparent output of the model

3.1 Dataset Description

We used 12 features of 1,205 clinical records of the Maternal Health and High-Risk Pregnancy dataset to implement strict preprocessing strategies (imputation, label encoding, and standardization). We considered five foundational ML algorithms, which are Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Xtreme Gradient Boosting (XGBoost), and Decision Tree (DT). To optimize the predictive stability, a Soft Voting Ensemble consisting of SVM, KNN, DT, and XGBoost were created. All of the models were tested with Stratified 5-Fold Cross-Validation. SHapley additive explanations (SHAP) were applied to understand feature importance and solve the black box issue of the conventional ML:

- **Demographic:** Age
- **Physiological:** Systolic Blood Pressure (BP), Diastolic BP, Blood Glucose (BS), Body Temperature, Heart Rate.
- **Clinical History:** BMI (Body Mass Index), Prior Complications, Diabetes Preexisting, Gestational Diabetes, Mental Health.
- **Target:** Risk Level (Coded as Low and High).

The dataset will especially be useful in the current research since it involves real-life variation in maternal health outcomes that may be characteristic of resource-limited healthcare environments.

3.2 Data Exploratory Analysis (EDA)

Prior to training machine learning algorithms, there was a thorough Exploratory Data Analysis (EDA) to learn the statistical characteristics and distribution of the dataset. The descriptive statistics of the numerical features are given in the Table 1. There is a total of 1,205 records in the dataset though the count row shows that some of the features are missing in some cases with most of them being BMI (1,187 records) and Systolic BP (1,200 records). This requires an imputation method on the preprocessing stage to maintain the integrity of data.

Table 1 Descriptive Statistics of Clinical Features

Feature	Count	Mean	Std	Min	25%	50%	75%	Max
Age	1205.0	29.8	13.5	10.0	19.0	26.0	39.0	70.0
Systolic BP	1200.0	113.2	18.4	70.0	100.0	120.0	120.0	160.0
Diastolic BP	1201.0	76.5	13.9	49.0	65.0	80.0	90.0	100.0
BS (Glucose)	1203.0	8.7	3.3	6.0	6.9	7.5	8.0	19.0
Body Temp	1205.0	98.7	1.4	98.0	98.0	98.0	98.0	103.0
BMI	1187.0	20.1	5.6	15.0	17.0	21.0	23.0	35.0
Heart Rate	1203.0	74.3	8.1	70.0	70.0	76.0	80.0	90.0

In an effort to identify the balance of data, we examined the balance of target variable, RiskLevel. Figure 2 shows the ratio between Low and High-Risk cases.

Moreover, Figure 3 shows a Correlation Heatmap, which demonstrates association with physiological attributes and the target variable. Significant relationships were found between Systolic BP, Diastolic BP, Blood Glucose (BS), and Age with an increased risk, which makes preliminary statistical evidence of the feature importance

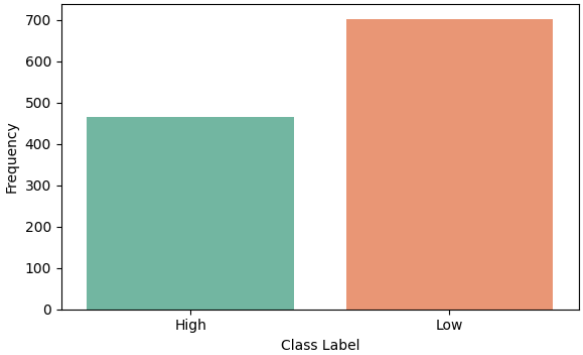

Fig. 2 Distribution of the Target Variable (Risk Level). The dataset exhibits a class imbalance, which supports the use of stratified cross-validation to ensure reliable model evaluation.

Figure 4 shows the frequency of the numeric characteristics that were used in this research to make predictions about high risk pregnancy. The visualization gives information about the distribution of data, central tendencies, and the possible imbalance in classes in terms of clinical indicators. The age distribution indicates that the majority of pregnant women are between the 20-35 years bracket that is regarded as the average reproductive age, and the other minor percentage of the cases are those of advanced maternal age after 35 years. The systolic and diastolic blood pressure values are clustered in the clinically normal to slightly elevated ranges, nonetheless, there are some outliers, which suggest hypertensive ailments, which can cause pregnancy complications. The distributions of the blood glucose levels (BS) are right skewed with the majority of the observations being within the normal ranges with significant higher values indicating the possibility of the diabetic conditions. Body temperature is very concentrated around normal values of physiological values, meaning that there is a slight variation and this proves the consistency of the data. Conversely, body mass index (BMI) presents more scatter plot with a high percentage of those found in the category of overweight and obese which are recognized risk factors of poor pregnancy outcomes. Binary clinical variables including past complications, pre-existing diabetes, gestational diabetes, and mental health status indicate the imbalance in the classes, which mostly belong to the negative one. This bias brings about the essence of applying strong modeling methods and validation mechanisms in order to prevent biased forecasts. The values of heart rate are mostly within the normal physiological values of the heart rate with slight variations which could be indicative of personal health conditions.

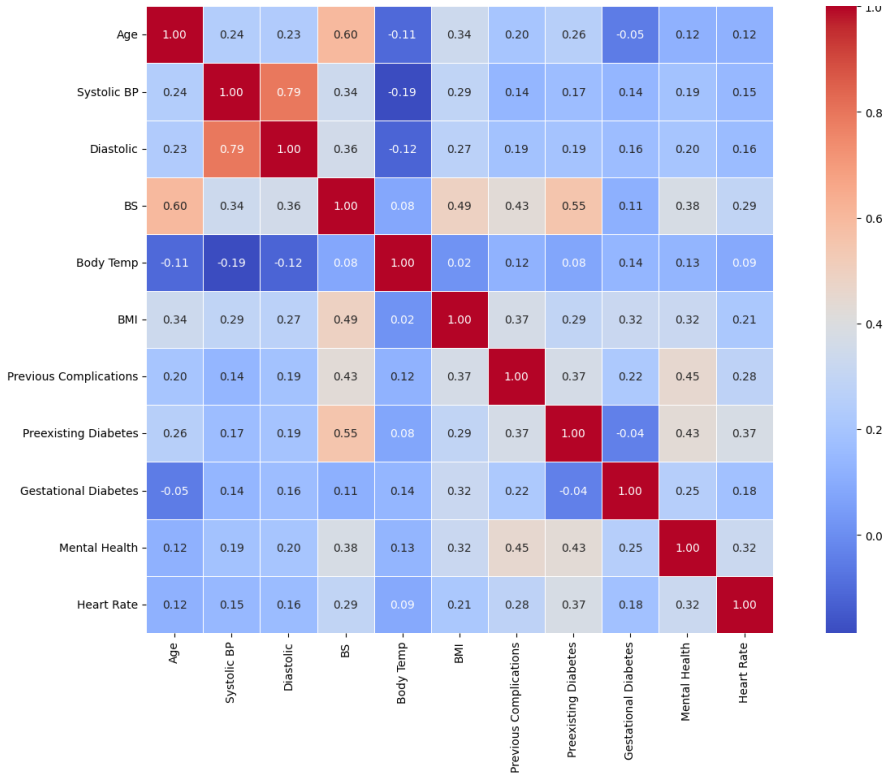

Fig. 3 Pearson Correlation Heatmap. Lighter/Warmer colors indicate stronger positive correlations, highlighting the physiological factors most strongly associated with pregnancy risk.

3.3 Data Preprocessing

The raw data went through various stages of preprocessing to provide the reliability of the machine learning models:

1. **Missing Value Tolerance:** The data was checked on missing values. Although most of the records were complete, minor data issues such as BMI were resolved through the median imputation so that the data distribution integrity would not suffer at the expense of dropping useful samples.
2. **Label Encoding:** The risk level variable is nominal. We used label encoding in order to encode the classes into numbers: 0 (Low Risk), 1 (High Risk).
3. **Standardization:** The Age, Blood Pressure, and Heart rate are features with a considerable scale difference. To make sure that distance-based algorithms (such as KNN and SVM) are not biased to the feature with big values, we used a Standard Scaler in normalizing the continuous variables to the mean of 0 and standard deviation of 1.

3.4 Machine Learning Algorithms

We used five different base classifiers to have a baseline performance of the classifier:

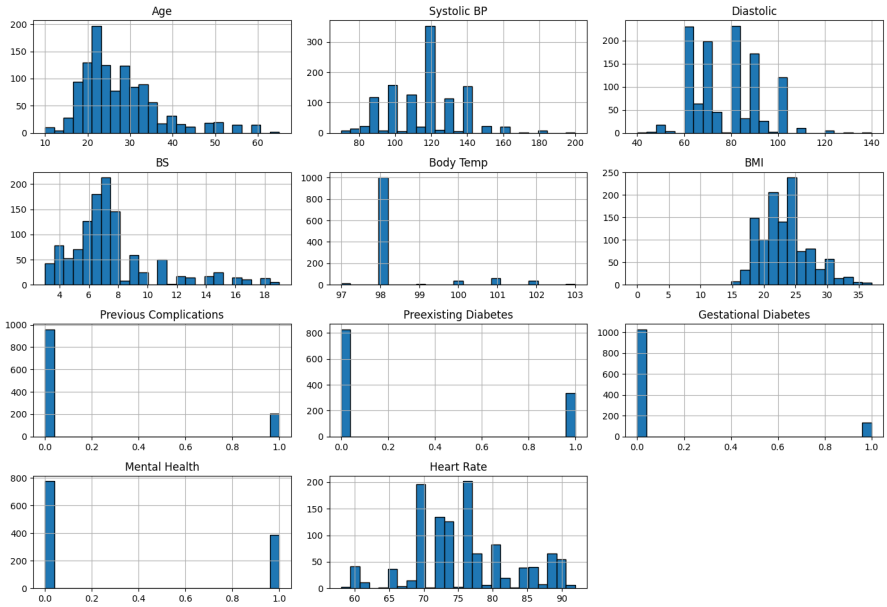

Fig. 4 Distribution of numerical maternal health features, including physiological measures and clinical indicators. Most values fall within normal ranges, while variability in BMI, blood glucose, and blood pressure highlights key risk factors and class imbalance relevant to high risk pregnancy prediction.

- **Random Forest (RF):** This is a group of decision trees that decreases overfitting using bagging.
- **Support Vector Machine (SVM):** This is an effective classifier that is useful in high-dimensional space with the radial basis function (RBF) kernel used herein.
- **K-Nearest Neighbors (KNN):** It is a non-parametric approach that groups the examples of similar features.
- **Xtreme Gradient Boosting (XGBoost):** A very fast and useful implementation of gradient boosted decision trees which is famous because of its speed and capability on structured data.
- **Decision Tree (DT):** An easy-to-understand model, simple and can be interpreted to split data in various groups depending on the feature values, which is used as a basic block to make up an ensemble.

3.5 Proposed Soft Voting Ensemble

In order to stabilize our model and generalize on unknown data, we formed a Soft Voting Ensemble. Figure 5 represents the architecture flow of this proposed framework. The workflow will take the following stages as illustrated by the figure:

- **Input Layer:** It is the processed feature vector or 12 attributes that is the input to all base learners at a time.
- **Parallel Model Running:** Four different classifiers SVM, KNN, Decision Tree, and XGBoost are applied to the input information independently. These particular

algorithms were chosen to ensure that there is a wide range of different algorithms: SVM tries to find the best hyperplanes, KNN is based on the idea of the similarity of instances, Decision Trees use rule-based splits, and XGBoost is a gradient-boosted refinement of decisions.

- **Probability Generation:** Each classifier i in lieu of producing a single label (Hard Voting) production computes a probability distribution $P_i = [p_{\text{low}},\ p_{\text{high}}]$ indicating how confident the classifier is with each risk level.
- **Soft Voting Mechanism:** The ensemble uses the weighted average probability over each of these classes j of all of the N classifiers.
- **Final Decision:** The system that has the highest average probability (Argmax) of producing the final prediction $\hat{y}$ is the one that is identified.

$$\hat{y} = \arg\max_{j} \left(\frac{1}{N} \sum_{i=1}^{N} P_{ij} \right) \tag{1}$$

Subtle nuances of the data are captured by this probability-averaging technique that enables the ensemble to do so. As an example, when one model is a little more uncertain (i.e. 40 percent High Risk) and others are very certain (i.e. 90 percent High Risk), the certainty of the others will offset the uncertainty, creating a more consistent and reliable prediction than either one model would have made on its own.

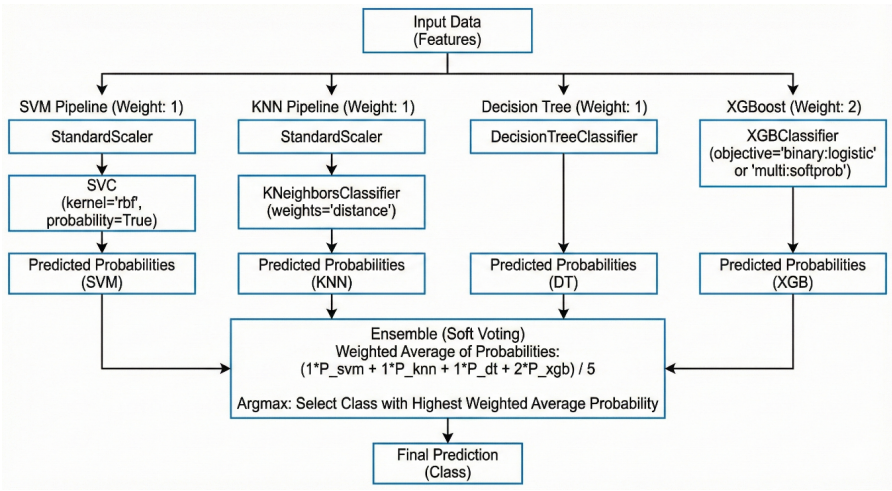

Fig. 5 Schematic diagram of the proposed Soft Voting Ensemble Framework.

3.6 Model Evaluation and Validation

In order to be certain that the derived findings are statistically sound, rather than chances of a good data division, Stratified 5-Fold Cross-Validastion is used in this study. In this validation strategy, the dataset is partitioned into five subsets, which are mutually exclusive, and the original distribution of classes of High and Low-risk is

maintained in each of the folds. Class balance is especially significant in medical data where the class imbalance may affect the performance of the model.

In the validation process the model is trained using four folds and tested on the rest of the fold. This is done five times with each subset acting as the test set once only. The final performance is calculated as the average of the performance of the model on all folds, thus, having a stronger estimate of the ability of the model to generalize.

In order to assess predictive performance and stability of the models the following statistical measures are reported:

- **Measures:** We present the Mean Accuracy of the 5 folds and Standard Deviation (SD).
- **Significance:** The smaller the SD, the more consistent and valid model, which is one of the main goals of the present research.

$$SD = \sqrt{\frac{1}{K}\sum_{k=1}^{K}(Acc_k - \overline{Acc})^2} \tag{2}$$

Where, $\overline{Acc}$ = mean accuracy

3.7 Explainability Framework (SHAP)

We combined SHapley Additive exPlanations (SHAP) in order to achieve clinical transparency. SHAP values statistically measure the percentage of the total prediction of the model to features and thus, the result is the overall output of the model being summed up as the feature effects. Using TreeExplainer on tree-based models (such as XGBoost and Random Forest) makes it possible to compute the precise Shapley values, so that a mathematically coherent explanation of why a particular prediction was made can be obtained. This enables a shift in predictions of black-boxes to white-box, which is paramount to medical AI.

4 Results and Discussion

In this part, the performance of the adopted machine learning models and the interpretation of the results on SHAP are displayed. Stratified 5-Fold Cross-Validation is used to provide the results with reliability.

4.1 Comparative Model Performance

A summary of the five base classifier performance and the proposed Soft Voting Ensemble is seen in Table 2. The two major measures used are Mean Accuracy and Standard Deviation(SD).

As it is seen in Table 2, XGBoost provided the highest raw accuracy (98.89%). The Proposed Soft Voting Ensemble was, however, found to be critically stable as the standard deviation was lowest at 0.32% as opposed to XGBoost at 0.58 and KNN at 1.31.

To further examine the robustness of these models, Table 3 presents the accuracy of each of the classifiers in each of the cross-validation folds. Reliability is in many

Table 2 Performance Comparison of Classifiers (Stratified 5-Fold CV)

Model	Mean Accuracy	Standard Deviation	Stability Assessment
XGBoost	98.89%	0.58%	High Acc., Moderate Stability
Proposed Ensemble	98.80%	0.32%	Highest Stability
Random Forest	98.63%	0.50%	High Accuracy
Decision Tree	97.60%	0.88%	Moderate
SVM	97.43%	0.47%	Moderate
KNN	96.91%	1.31%	Low Stability

cases favored in a clinical context when compared to the marginal gain in accuracy. Ensemble model also allows prediction quality to be similar to that in other patient target groups which is the guarantee that overfitting to a particular pattern of data is avoided.

Table 3 Fold-wise Accuracy Breakdown

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean
XGBoost	97.86%	99.57%	99.14%	98.71%	99.14%	98.89%
Ensemble	98.29%	98.71%	99.14%	98.71%	99.14%	98.80%
Random Forest	97.86%	98.28%	99.14%	98.71%	99.14%	98.63%
Decision Tree	97.86%	98.71%	96.57%	96.57%	98.28%	97.60%
SVM	97.44%	96.57%	97.42%	97.85%	97.85%	97.43%
KNN	95.73%	96.14%	98.28%	95.71%	98.71%	96.91%

4.2 Classification Efficacy of the Ensemble Model

To provide a granular view of the proposed Soft Voting Ensemble's performance, we analyzed the Confusion Matrix and Receiver Operating Characteristic (ROC) Curve. The Confusion Matrix of the Ensemble classifier is shown in Figure 6. The diagonal components are correct predictions and those that are off the diagonal represent misclassifications

Figure 7 shows the ROC Curve and the Score of Area under Curve (AUC) of each of the classes.

4.3 Comprehensive Evaluation of the Ensemble Model

In order to strictly test the suggested Soft Voting Ensemble on the underlying class imbalance of the data, we were testing Precision, Recall, and F1-score and the overall accuracy illustrated in Table 4.

The ensemble model had a very high level of performance in all diagnostic metrics. In medical diagnostic systems, Recall (Sensitivity) is commonly taken to be the most important measure since the cost of a false negative (a risk not detected) is much greater than that of a false positive. The ensemble has an extremely high recall of

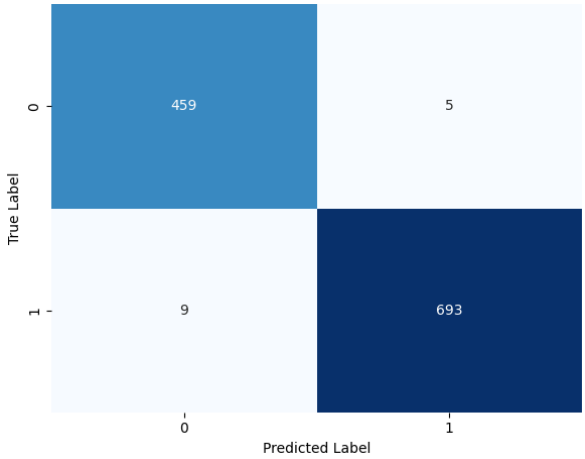

Fig. 6 Confusion Matrix of the Soft Voting Ensemble. The high values along the diagonal indicate strong classification performance across all three risk levels (Low, High). The minimal misclassification rates demonstrate the model's ability to effectively distinguish between class boundaries.

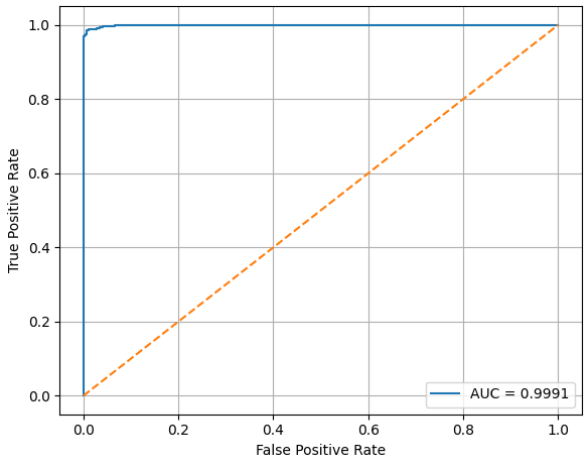

Fig. 7 Receiver Operating Characteristic (ROC) Curve for the Ensemble Model. The curves for all three classes (Low, High) approach the top-left corner, with AUC values nearing 1.0. This confirms the model's high sensitivity and specificity, indicating it can reliably discriminate between risk levels without significant false positives or false negatives.

0.99, both on the two classes, which means that the ensemble has a strong ability to find all the relevant instances of the data correctly and without missing any cases.

Moreover, the model had a high Precision (0.98 Class 0 and 0.99 Class 1) meaning that at the selected risk level that the framework has indicated, the prediction is quite accurate hence avoiding unnecessary clinical interventions (false positives). F1-Score which gives the harmonic mean of precision and recall was also at 0.98 and 0.99 respectively. The fact that the macro and weighted averages are no less than 0.99 is a confirmation that the Soft Voting Ensemble is effective in countering bias against the

Table 4 Overall Classification Report for the Soft Voting Ensemble

	Class	Precision	Recall	F1 Score	Support
	0	0.98	0.99	0.98	464
	1	0.99	0.99	0.99	702
Accuracy				0.99	1166
Macro avg		0.99	0.99	0.99	1166
Weighted avg		0.99	0.99	0.99	1166

majority class (Class 1, support: 702), indicating it is stable and has an overall high effectiveness with the unbalanced dataset

4.4 Interpretability Analysis (SHAP Results)

In an attempt to confirm the clinical viability of the strong-performing models, we used SHAP analysis. The following section highlights the description of the 'Why' behind the predictions.

Figure 8 illustrates a SHAP Waterfall Plot that explains the details of a particular person who is classified as being in a 'High Risk' category.

In Figure 8, the final prediction f(x)=1.00 (High Risk) is driven by specific patient values.

- **Positive Drivers (Red):** The Age and Systolic BP of the patient all play a significant role to increase the risk score (+0.11 respectively). This is in tandem with medical consensus that advanced maternal age and hypertension are best pointers against risks.
- **Base Value:** This is cumulative to a certainty (1.00) where the prediction begins with an average probability of risk (0.507).

Although the waterfall plot describes the cases in particular cases, Figure 8 shows the risk factors identified by the model throughout the entire population all over the world.

Figure 9 reflects that the best feature is Blood Glucose (BS). These red dots (high glucose values) are greatly concentrated on the right (positive SHAP values) which means that the high level of blood sugar is the most influential factor to predict the high-risk pregnancy. This confirms the importance of controlling diabetes among the maternal health. Just like glucose, the higher the BMI values (red) the more the risk. On the contrary, smaller BMI (blue) has the propensity of reducing the risk prediction. The age distribution is broader indicating that age is a risk factor but its effect differs more among patients than the glucose levels.

These results support the notion that it is not merely pattern-matching noise that these results are indeed reflecting the presence of the right physiological drivers (Diabetes, Hypertension, Obesity).

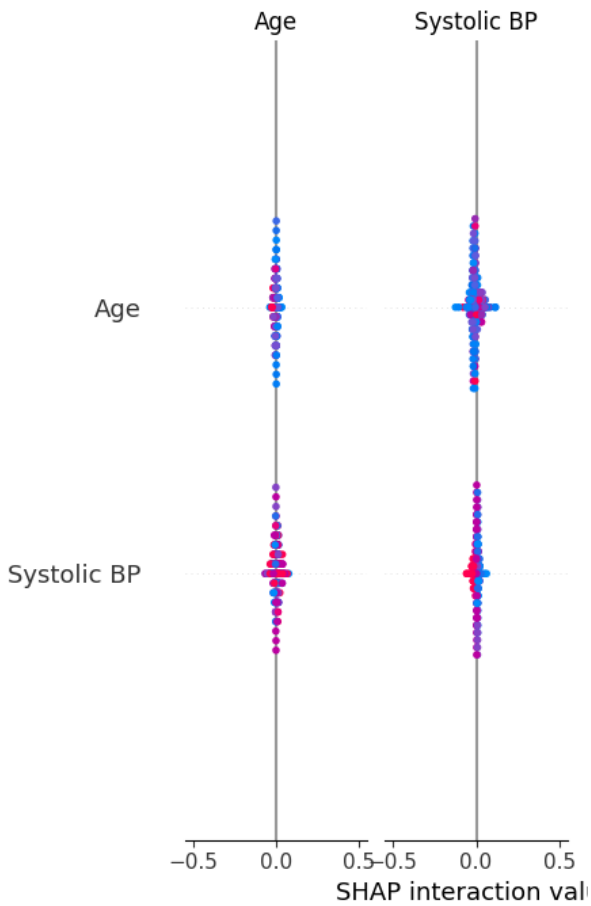

Fig. 8 SHAP Waterfall Plot for a single high-risk instance. The plot visualizes how specific feature values push the prediction from the base value $E[f(x)]$ towards the final model output $f(x)$.

5 Conclusion

This research study dealt with the key issue, which is automating identification of high-risk pregnancies in medical institutions under resource constraint. Using a set of 1,205 clinical records, we constructed and tested an effective machine learning system that does not only have a high predictive performance but also has a high degree of clinical reliability and interpretability. The experimental findings emphasized an important observation on the trade-off between the accuracy and the stability. Although xgboost model had the best individual mean accuracy of 98.89, it had a larger standard deviation (0.58) when compared to the cross-validation folds. Contrary to that, the suggested Soft Voting Ensemble (the combination of SVM, KNN, DT, and XGBoost) had an identical accuracy of 98.80% but more stability with the minimum possible standard deviation of 0.32. In medical diagnostics where consistency is the most crucial aspect, this reduction in variance helps the Ensemble to be a more valid choice to be

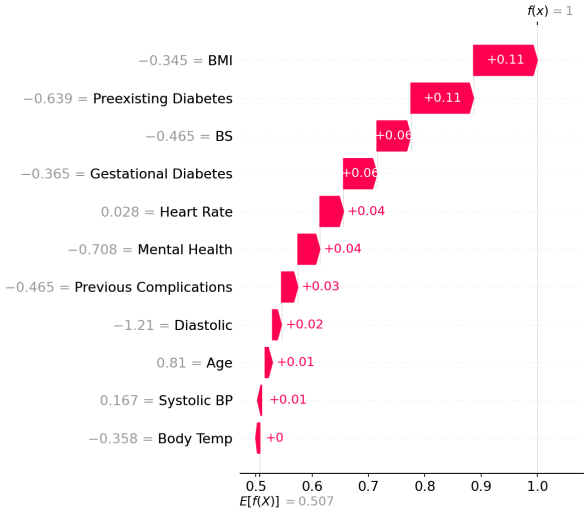

Fig. 9 SHAP Summary Plot (Beeswarm) illustrating the global impact of features on model predictions. Features are ranked by importance from top to bottom.

implemented in reality. Moreover, it was successful in filling the black-box gap with the integration of SHAP (SHapley Additive exPlanations). The correlation analysis has verified that all the models are based on physiologically reasonable indicators, namely Blood Glucose, BMI, and Systolic Blood Pressure; instead of accidental correlations. This visibility will be necessary to gain trust among obstetricians so that they can take the AI system as a proven second opinion and not as an opaque decision-maker. Overall, the study offers a very stable, predictable, and precise computing instrument regarding the health of mothers, which can be used to help intervene at an early age and probably diminish the rate of maternal mortality in developing countries.

6 Future Work

Although some positive findings were achieved by today's research, a number of future research opportunities still exist:

1. **Expansion and Diversity of Data:** The existing model was conditioned on the data mainly found in rural Bangladesh. Confirmation of the framework using various datasets across the geographical and demographical populations would involve a greater generalizability of the findings across all global locations.
2. **Real-Time IoT Implementation:** Subsequent versions may include implementing the Soft Voting Ensemble at the edge (e.g., smart wearables) to detect vital signs in real-time and conduct and assess the risk of expectant mothers in remote locations.
3. **Hybrid Deep Learning Models:** Hybrid model combinations of structured data processing of ensemble trees with the feature discovery attributes of Deep Neural Networks (DNNs) may also yield better performance, specifically non-linear risk patterns that are more difficult to forecast.

4. **Longitudinal Study:** A longitudinal study that follows-up the patients during their pregnancy would permit creating dynamic risk prediction models adapting during the course of the pregnancy and providing more personalized care.

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Declarations

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Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The dataset used in this study is publicly available. The maternal health records were obtained from the *Maternal Health and High-Risk Pregnancy Dataset* available on Kaggle, contributed by V. Mohammed Raiyyan, and originally hosted in the Mendeley Data repository.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Code availability

The source code used in this study is not publicly available at the time of publication. However, it will be made available by the corresponding author upon reasonable request.

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